DO NOW — RECALL

Lesson 112 · Choosing Applications and Utility Software

Name: _____ Class: _____ Date: _____

1. What role does a device driver perform?

2. A linked-list node has data and what other essential field?

3. In a BST, where is a value less than the current node placed?

4. Why can an array provide direct indexed access?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It lets the OS communicate with/control a specific hardware device.

2. Model answer

A link/reference to the next node (for a singly linked list).

3. Model answer

In its left subtree (under the chosen no-duplicate convention).

4. Model answer

Its element locations can be calculated from base address and index.

By the end, you can…

• Identify users, scale, accessibility, platform and constraints.

• Select task-specific tools: DBMS, spreadsheet, CAD, DTP, etc.

• Backup, compression, disk tools, antivirus and encryption maintain/protect systems.

Vocabulary — know these terms

application software — Programs that meet a user task.

utility — Software for maintenance, protection or optimisation.

DBMS — Database management system.

backup — Recoverable copy to protect against data loss.

NEW LEARNING

Read, annotate and explain

Clarify needs

Identify users, scale, accessibility, platform and constraints.

Match applications

Select task-specific tools: DBMS, spreadsheet, CAD, DTP, etc.

Understand utilities

Backup, compression, disk tools, antivirus and encryption maintain/protect systems.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

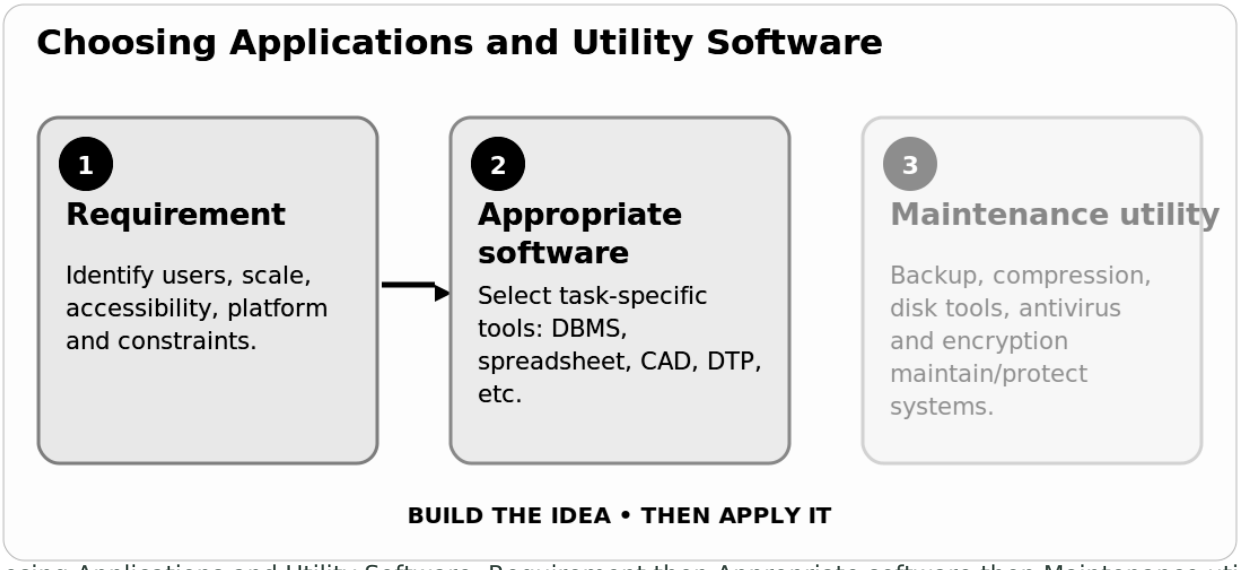
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Choosing Applications and Utility Software: Requirement then Appropriate software then Maintenance utility

New Learning

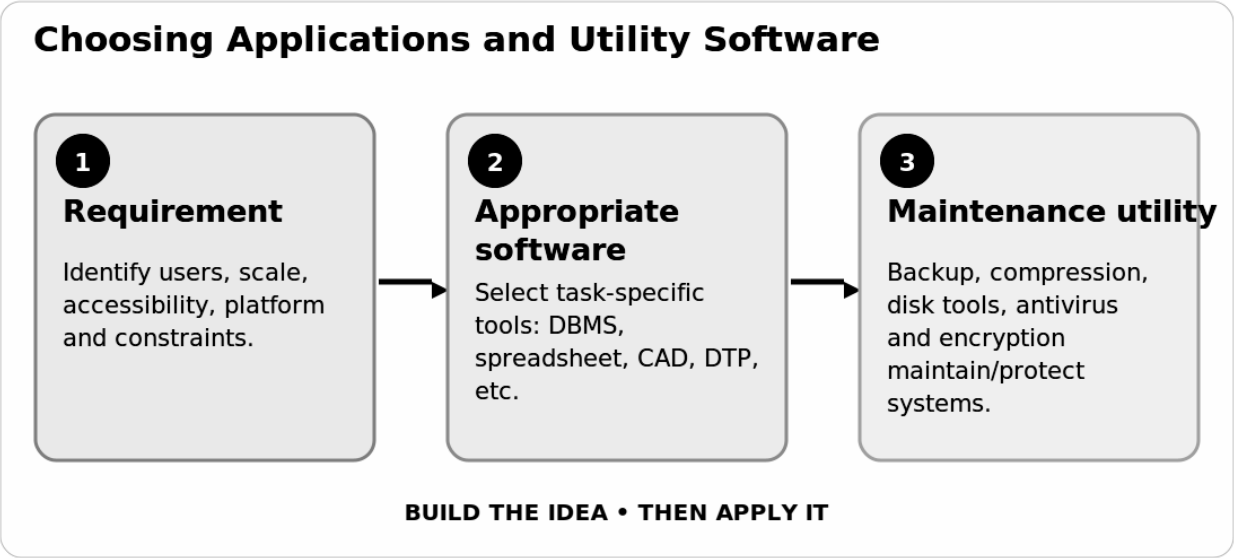


Choosing Applications and Utility Software: Requirement then Appropriate software then Maintenance utility

New Learning · original lesson diagrams

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New Learning



Choosing Applications and Utility Software: Requirement then Appropriate software then Maintenance utility

I DO — WORKED EXAMPLE

Follow the worked method

A clinic manages thousands of appointments accessed concurrently. Spreadsheet or database application?

Model answer

A DBMS is better: structured records, querying, concurrency, constraints and access rights. A spreadsheet is useful for small-scale analysis but is not ideal as a multi-user transactional store.

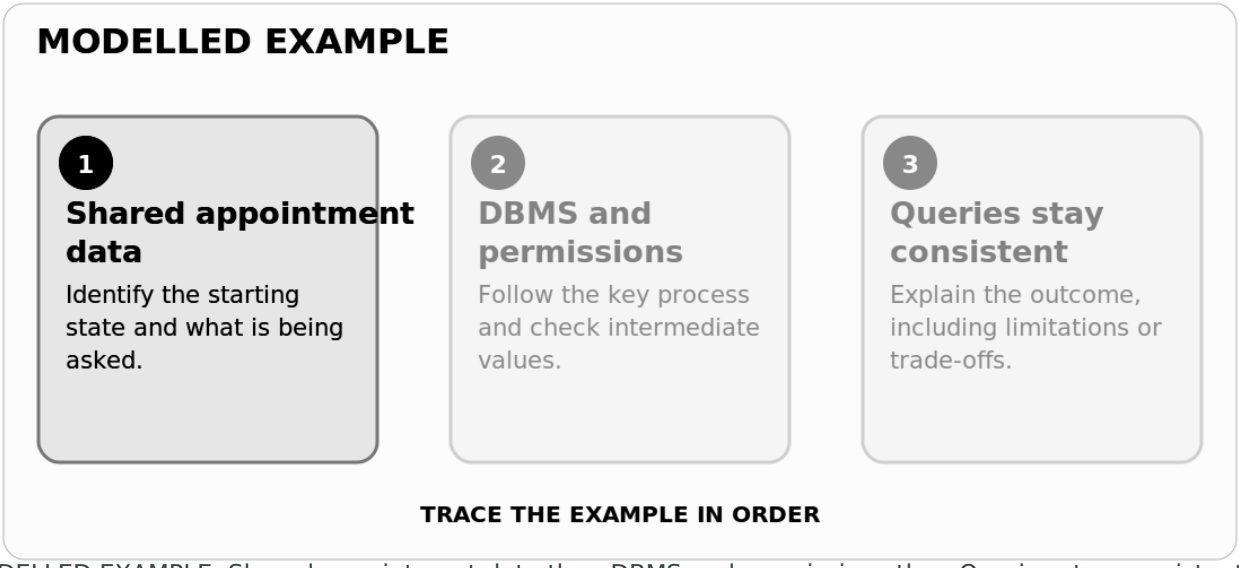
Check: link each claim to the scenario and use precise OCR terminology.

I Do · original lesson diagrams

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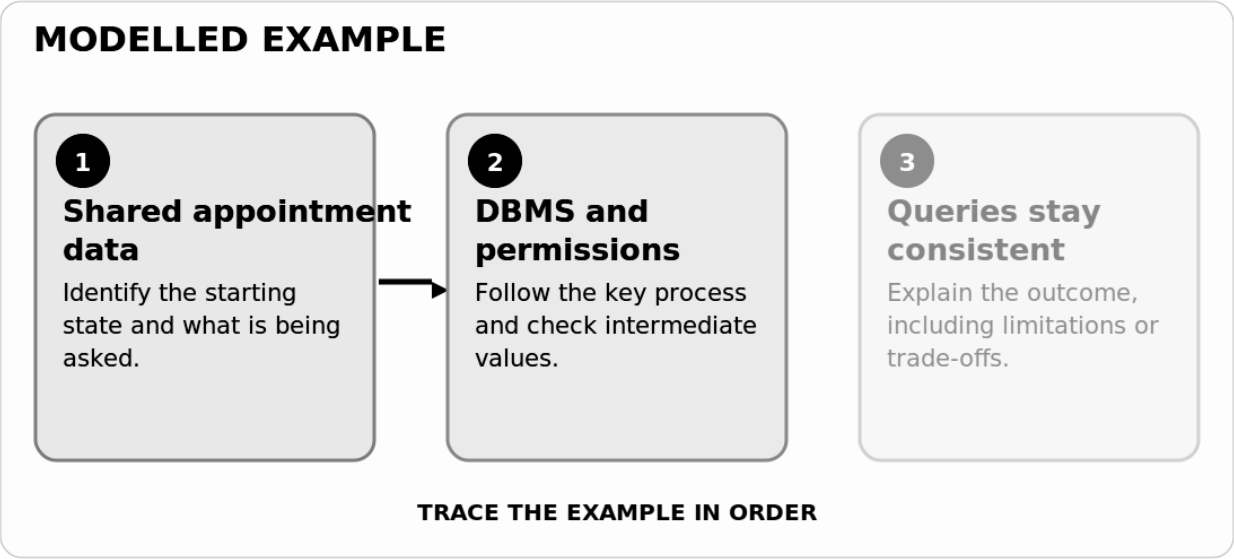
MODELLED EXAMPLE: Shared appointment data then DBMS and permissions then Queries stay consistent

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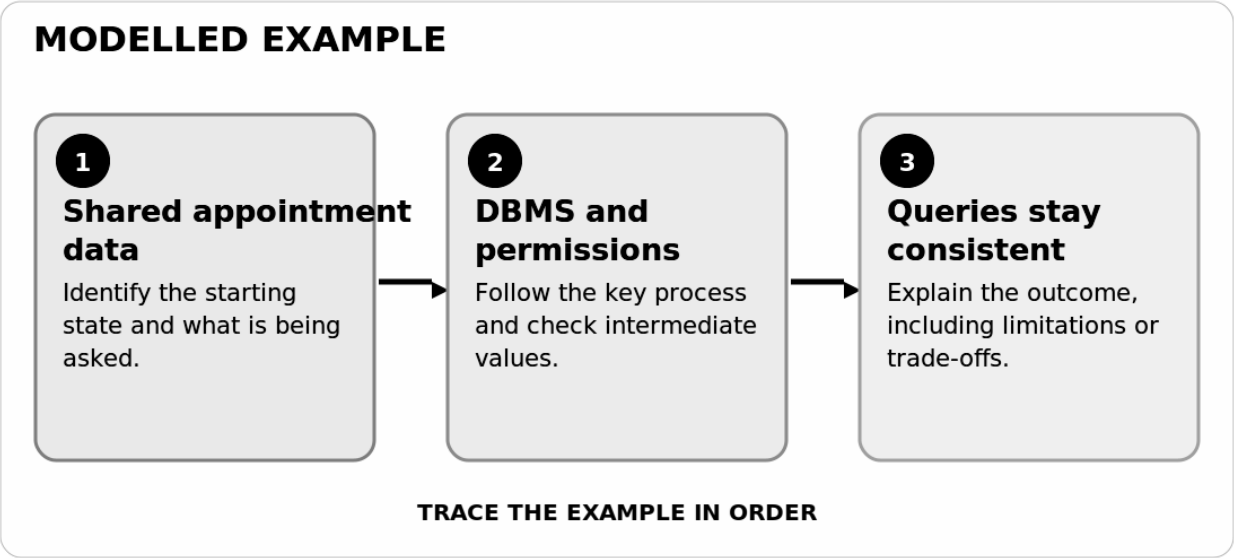
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WE DO — GUIDED PRACTICE

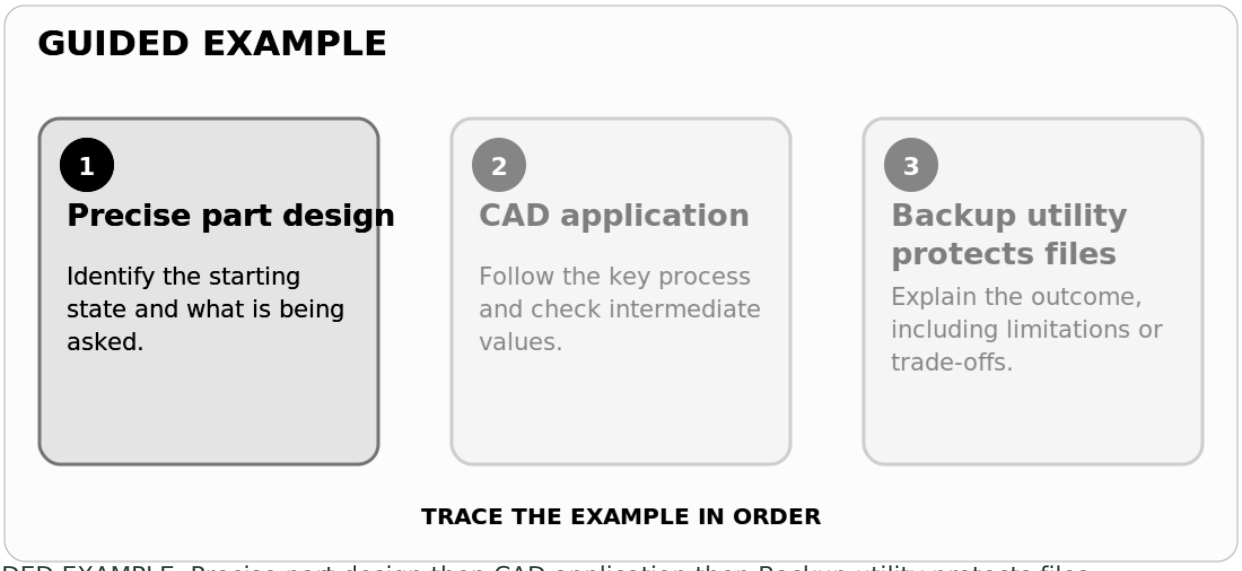
Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A designer produces a precise engineered component and needs automated backups. Which software types?

Identify the inputs, relevant facts and exact output or judgement required.



GUIDED EXAMPLE: Precise part design then CAD application then Backup utility protects files

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: What must we decide? Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Justify an application for editing a school magazine with text and complex page layout. [6]

Question 2

Differentiate application software from utility software using two examples. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A small charity needs inventory records, backups and accessible user interfaces. Recommend software and justify limitations. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A backup utility and a word processor are both examples of utility software.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

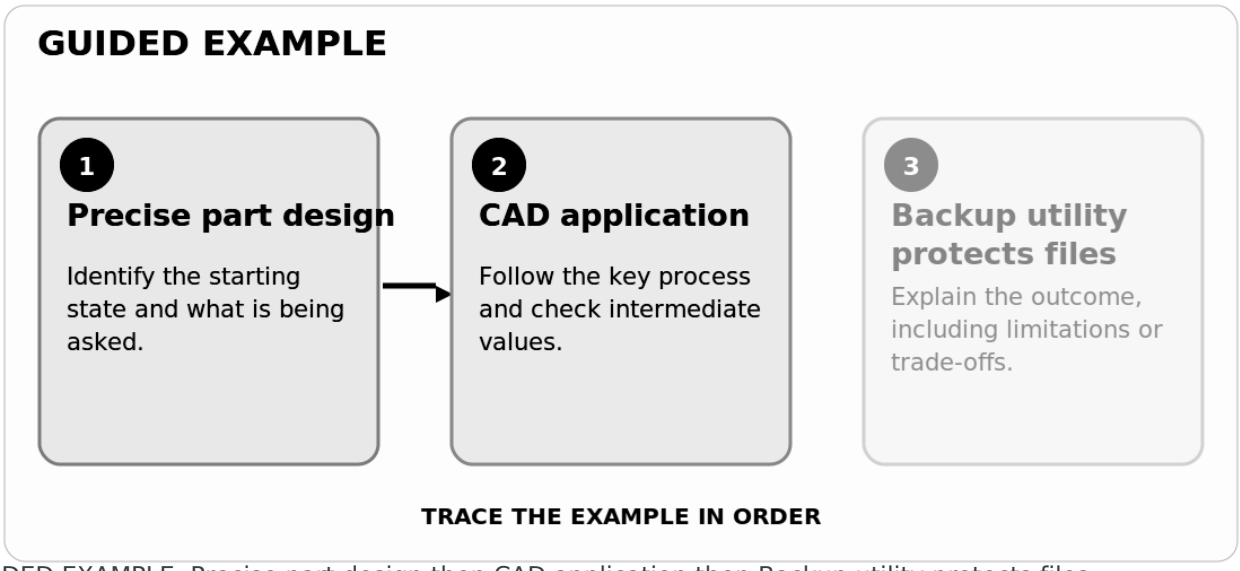
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

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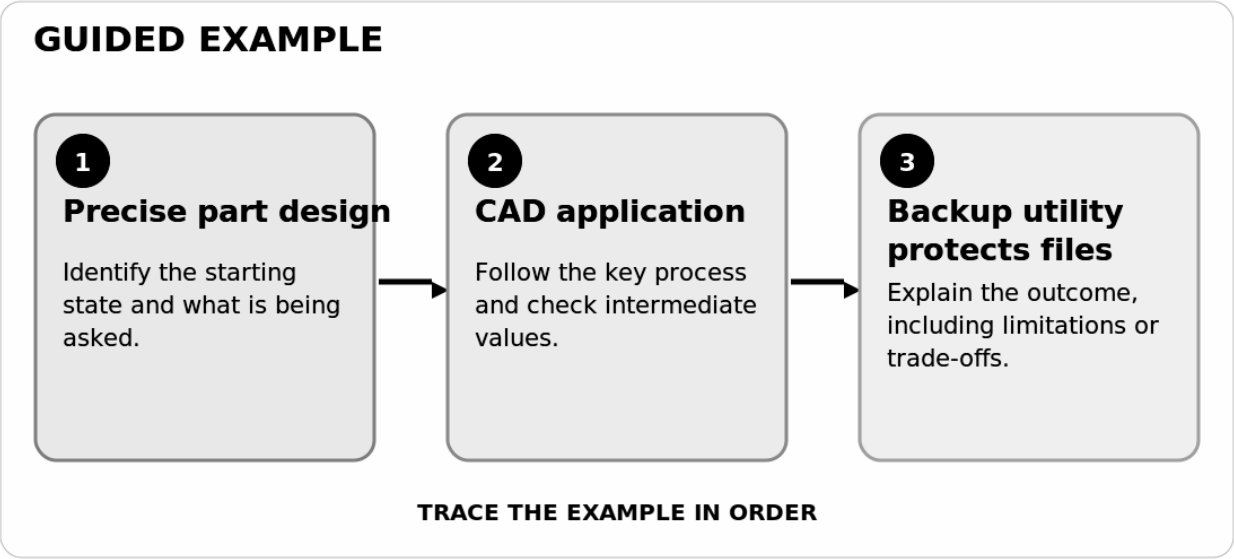
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WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Work through the method

Use the middle stage of the visual to follow the process and state each intermediate result.

Shared worked answer

CAD suits accurate technical modelling with dimensions and revision. A backup utility produces recoverable copies; design software and utility software have different purposes.

IMPROVE: Explain one correction to your own reasoning.